

Machine Learning Design

Service Failure / SLA Risk Prediction

1. ML objective

Predict whether a service is likely to experience a material incident within a defined future window, such as the next 30 minutes.

2. Target

```
incident_next_30m ∈ {0,1}
```

3. Baseline and candidate models

- Logistic Regression — interpretable baseline.
- Random Forest — nonlinear interactions.
- Gradient boosting/XGBoost — stronger candidate for tabular data.

4. Training pipeline

```
Historical telemetry
↓
Data validation
↓
Temporal split
↓
Feature engineering
↓
Train
↓
Validation
↓
Model selection
↓
Calibration
↓
Model artifact + version
```

5. Leakage prevention

- Never use telemetry from after the prediction timestamp.
- Generate labels using future windows.
- Use time-based validation instead of random-only splitting for temporal data.
- Fit preprocessing only on training data.

6. Metrics

- Recall for critical incidents.
- Precision to control alert fatigue.
- F1 for balance.
- PR-AUC for imbalanced events.
- Calibration error/Brier score when probabilities are used operationally.

7. Production inference

```
Telemetry window
↓
```

```
Feature builder
↓
Versioned preprocessing
↓
Model
↓
risk_score
↓
risk_level
↓
PostgreSQL risk_assessment
```

8. Monitoring

- Feature drift.
- Label drift.
- Prediction distribution.
- False-negative rate after labels mature.
- Model version comparison.